

Network Devices

Learning Objectives

- Identify common network devices and their functions
- Explain the role of routers, switches, and modems
- Differentiate between wired and wireless devices

Engage (5-7 minutes)

Picture a railway junction. Trains arrive from different cities, and the station master directs each train to the correct platform. Network devices work the same way—they receive data and route it to the right destination. Today we meet the station masters of the computer world.

Explore (10-12 minutes)

Look at the networking corner of your computer lab. Can you find a modem, router, or switch? Draw each device you see and write one sentence describing what you think it does. Compare your drawings with a classmate.

Explain (10-12 minutes)

A modem converts data between your network and the internet. A router directs data packets to the correct devices on a network, like a postmaster sorting letters. A switch connects multiple computers within a LAN so they can communicate. A hub is an older device that sends data to all ports regardless of destination.

Elaborate (8-10 minutes)

Think of the bus stand in your town. The modem is like the bus that travels between towns carrying passengers. The router is the bus stand conductor who directs passengers to the right bus. The switch is like the seating arrangement inside the bus that lets passengers talk to their neighbors. Which device would you need to connect two computers in a small office?

Evaluate (5-8 minutes)

Exit ticket: Name three network devices and write one function for each. If you finish early, explain why a switch is better than a hub for a school network.